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Vouch

Vouch checks a resume against a job post, one requirement at a time.

For each thing the post asks for, it finds the line of the resume that proves it. If there is no such line, it says so.

THE PROBLEM

THE POST

A job post asks for twenty-odd separate things.

THE READER

A person reads it once and forms an impression.

THE ANSWER

"I think I am a good fit." Nothing was checked item by item.

Nobody checks a job post. They read it once and guess.

I built this because I am job hunting. The demo runs on your job post.

WHY THE OBVIOUS BUILD IS WRONG

THE BUILD	One model call. The post and the resume go in together.
WHAT IT SAYS	"Yes. Strong match on Kubernetes."
THE PROBLEM	The resume may not say Kubernetes anywhere.

One model call answers.
Nothing shows where it came
from.

You cannot tell if the resume said it, or the model
made it up. Everything else in this project follows
from that one sentence.

HOW THE ANSWER IS MADE

THE RESUME	Never pasted into the prompt. It is split into numbered lines, and the agent searches it.
THE MODEL	Sees the six lines it asked for. Not a document it can drift over.
WHAT IT RETURNS	A line number and a verdict. No text of its own.
THE QUOTE	Read back out of the file at that number, after the model is finished.

So the model never writes the answer. It returns a line number.

The model has no way to write a quote. So it cannot invent one.

Decided before the code was written: decision 0001.

DEMO

FIRST	The extension, on a real job post. 45 seconds.
THEN	The page, for the rest of the five minutes.

Nothing on this slide is read out. The clicks are in `demo-script.md` .

READING THE FILE

A quote has to be a real line. So where the lines break is not a detail. It is the answer.
Two readers, same PDF: file order, or position on the page.

16 SEPTEMBER	LINES	EVIDENCED	PARTLY	THE CALL
pypdf	160	3	17	weak
pdfplumber	51	6	13	worth applying

So how the file is cut into lines
decides the answer.

The verdict changed. Nothing about the model
did.

Same resume, same post, same model. Runs minutes apart. Report 0020.

WHY THIS IS AN AGENT

It is an agent for three reasons.
Calling a model is not one of them.

IT HAS TOOLS

It writes its own search terms, one requirement at a time.

IT TAKES STEPS

It keeps state between them. The loop counts requirements, not model turns, so nothing stops early.

IT GOES BACK

Up to three searches, and it chooses when the first one found nothing.

WHY LANGGRAPH, AND NOT THE OTHER FOUR

LangGraph, because the audit goes back.

```
extract → search → judge → verify ┐→ retry →→ search
                                ↳ advance → next requirement → report
```

THE DECISION	An edge in the graph, not an <code>if</code> inside a node.
THE CHECK	<code>verify</code> always runs. It is not the model's call.
THE TRACE	LangGraph streams state node by node. The trace is the demo.
THE REST	FastAPI, <code>rank_bm25</code> , Groq — libraries. Not my code.

Over LangChain, CrewAI, Pydantic AI, LlamaIndex — and DSPy ReAct. Decision 0001.

WHAT IT GOT WRONG

THE OLD TRIGGER	Go round again when the model reports its own evidence as weak.
WHAT HAPPENED	Nothing. 0 of 21, 0 of 23, 0 of 21.
THE BAD PART	It stayed silent on verdicts that were wrong — where the model's own written reason said "does not mention".

The old retry asked the model to grade itself. It never fired.

Three real runs, 15 September. Decision 0007.

THE FIX

It fired three times. Nobody pressed anything.

THE NEW TRIGGER	A first pass that found nothing, or a quote that failed its check. Both are things I can see in the data.
WHAT IT MOVED	One answer went from not evidenced to evidenced, on a line that was checked.
WHAT I WATCHED FOR	No second pass talked itself into a match. Three stayed negative.

Same post, 15 September. Report 0005.

THREE MORE THINGS THAT ARE WRONG

RANKING	Retrieval used to rank by word count, so <code>python</code> beat <code>fastapi</code> . It reported a gap that was not a gap. Fixed with BM25 — <code>rank_bm25</code> , a library, not my code.
THE MIDDLE	About 60% of answers land in "partly evidenced". It absorbs both ends. Known, not solved.
NO BUDGET	Three searches per requirement is capped. How many requirements retry is not. One resume retried on 6 of 6.

All three were found by running it, not by reading it.

Reports 0004, 0021 and 0012.

WHAT I LEARNED

Almost every model problem turned out to be a data problem.

THE PDF

How the file was read moved the verdict from weak to worth applying. Slide 6.

RETRIEVAL

Better ranking closed a gap that was not a gap.

LINE BREAKS

Repairing them removed a false finding.

The model never changed. Not once. That is the thing I would carry into a customer deployment.

HOW IT WAS BUILT

I built this by directing agents, under a method I wrote first.

THE LOOP

Brief, work, check, report, stop. Every piece, every time.

THE GUARDRAILS

They refuse, not advise. A turn cannot end on a red check.

WHAT IT CAUGHT

18 of 22 reports name something the brief got wrong.

WHAT IT MISSED

The planning role offered to build, four times. I caught it, not the kit.

I briefed it, I read every report, I rejected work. docs/ is the handover: 26 briefs, 8 decisions, 24 reports.

FORMWORK.md, at the root.

WHAT THIS REALLY IS

This is not a resume tool. It audits a document against a rulebook.

THE RULEBOOK	A job post. Or a policy, a tender, a regulation.
THE DOCUMENT	A resume. Or a contract, a filing, a submission.
THE OUTPUT	One verdict per item, each carrying the line it came from — or it says there is none.

I have shipped this shape before: a bilingual compliance engine, auditing documents against an uploaded rulebook, for regulated customers.

WHY THAT IS COMMERCIAL

THE QUESTION	Every pilot dies on one sentence. "How do I know it did not make that up?"
PROMPTING ANSWERS IT	With a probability. The model is told to behave, and usually does.
THIS ANSWERS IT	With architecture. The model cannot write text at all.

"It cannot invent a citation" is what makes a model usable in a regulated industry.

A guarantee you can point at beats a number you have to trust.

SCALING IT, HONESTLY

WHAT ALREADY WORKS	One document, one graph run, nothing shared. It is parallel by construction.
THE BUSINESS CASE	A free local screen decides whether a paid call is worth making. At ten thousand documents, that is the whole argument.
WHAT IT DOES NOT HAVE	An evaluation set with gold answers. Measured token cost. Durable orchestration. A human review gate. Per-tenant isolation.

Make the guarantee structural,
not a promise.

Vouch refuses to claim what it cannot evidence.
The method refuses to let a turn end on a red
check.